

#	Facility (short name)	Location (city/area)	Main chemicals / products	Primary hazard class (for MVP)	Example emergency scenario you can model
1	IOCL Guwahati Refinery	Noonmati, Guwahati	LPG, naphtha, MS (petrol), ATF, SKO, HSD, LDO, petroleum coke, sulphur	Large-scale flammable liquids & gases; toxic sulphur compounds	Refinery tank-farm fire → toxic plume over Noonmati & downstream residential areas; pipeline leak into drains/Brahmaputra. iocl
2	Kamrup Industrial Gases Ltd (KIGL)	Maniram Dewan Rd, Noonmati, Guwahati	Oxygen, nitrogen, argon, acetylene, CO ₂ (cylinders & bulk)	Compressed/liquefied gases; oxidizers; flammable acetylene	Acetylene cylinder yard fire/explosion; oxygen cascade rupture causing intense fire in nearby workshops. scribd+1
3	Premier Cryogenics Ltd (Samaguri unit)	Gandheli Bebejia / Chandmari belt, Guwahati	Liquid nitrogen, liquid oxygen, other cryogenic gases	Cryogenic liquids; asphyxiation; oxidizer hazards	Cryogenic tanker rollover near Chandmari → rapid vaporization, asphyxiation risk in low-lying area; oxygen leak intensifying a road-accident fire. scribd
4	Assam Petro-Chemicals (methanol plant)	Namrup, Dibrugarh (with Guwahati office presence)	Methanol, formalin (formaldehyde solution), urea-formaldehyde resins	Flammable & toxic liquids; irritant/carcinogenic formaldehyde	Tanker crash leaking formalin into a river (similar to Jatinga incident) → water contamination & public health advisory; methanol storage fire with toxic fumes. kigl.co+1
5	BVFCL Namrup Fertilizer Complex	Namrup, Dibrugarh	Ammonia, urea (neem-coated), some organic fertilizers	Toxic, pressurized ammonia gas; high-pressure synthesis	Compressor-house fire with suspected ammonia leak (as in 2025 incident) → toxic plume over worker colony; ammonia storage rupture. kigl
6	BCPL (Assam Gas Cracker)	Lepetkata, Dibrugarh	Ethylene, propylene, LLDPE, HDPE, polypropylene, by-product gasoline	Highly flammable gases & monomers; polymer dust hazards	Cracker unit leak → vapor cloud explosion; polymer plant fire with smoke plume over Lepetkata & nearby villages.

7	Shiva Chemicals (representative acid/solvent unit)	Beltola / Guwahati (typical location for such traders)	Industrial acids (e.g., nitric/sulphuric), solvents, bleaching powder, cleaning chemicals	Corrosives; flammable solvents; oxidizing bleaching agents	Godown fire with acid/solvent spill → corrosive runoff into local drains; mixing of bleach with acids generating toxic chlorine gas.
8	Motii Chem Center (acid trader)	Dhirenpara, Guwahati	Nitric acid and other industrial acids	Strong corrosives; oxidizing acids	Drum rupture during unloading → acid spill on road, injuries to workers, corrosion of infrastructure, neutralization operation.
9	Chemical warehouse hub (generic 3PL)	Bamunimaidam / Kalapahar industrial belt, Guwahati	Mixed chemicals in drums/IBCs (solvents, acids, agro-chemicals, paints) for multiple clients	Mixed hazards; potential for cascading reactions	Warehouse fire involving multiple chemical consignments → complex smoke plume, runoff into storm drains, evacuation of nearby colonies.
10	Ellenbarrie / other industrial gas player (NE distribution hub)	Guwahati / nearby (as a representative large gas supplier node)	Oxygen, nitrogen, argon, CO ₂ , possibly acetylene (bulk & cylinders)	Compressed/liquefied gases; oxidizers	Bulk liquid oxygen tank leak causing oxygen-enriched atmosphere → rapid fire spread in adjacent plastic/packaging units.

How it affects the environment :

A. IOCL Guwahati Refinery (petroleum refinery)

Environmental effects:

- **Air:** Emissions of SO₂, NO_x, PM_{2.5}/PM₁₀, VOCs (including benzene, toluene, xylene), H₂S, CO, and greenhouse gases (CO₂, CH₄). [pmc.ncbi.nlm.nih](https://pubmed.ncbi.nlm.nih/)+2
- **Water:** Risk of hydrocarbon spills into drains/soil, oily wastewater, effluent with organics, heavy metals, sulphides. [scribd](#)
- **Soil & groundwater:** Leaks from pipelines/tanks can contaminate soil and shallow aquifers with petroleum hydrocarbons and BTEX compounds. [scribd](#)
- **Health impact zone:** Studies around refineries show elevated respiratory issues and cancer-related biomarkers within 5 km, with effects detectable up to 10–20 km downwind. [pmc.ncbi.nlm.nih](https://pubmed.ncbi.nlm.nih/)

B. Industrial gas plants (KIGL, Premier Cryogenics, other gas units)

Environmental effects:

- **Air:** In normal operation, mostly inert gases (N_2 , Ar) and oxygen; low toxic emissions.
- **Accident scenarios:**
 - **Acetylene / oxygen fires** → intense combustion, soot, CO, possible destruction of nearby structures.
 - **Cryogenic spills** → rapid vaporization, local oxygen enrichment or depletion, asphyxiation risk.
- **Soil/water:** Limited chemical contamination unless there's a major fire with burnt materials and firefighting runoff.

C. Methanol / formalin plant (Assam Petro-Chemicals, Namrup) & formalin tanker incidents

Environmental effects:

- **Air:** Methanol vapors (flammable, toxic), formaldehyde (irritant, carcinogenic).
- **Water:** Formalin spills into rivers (like the **Jatinga River incident**) can be toxic to fish and aquatic life, degrade water quality, and force bans on water use for drinking, bathing, irrigation.indiatodayne+2
- **Soil:** Spills on land can infiltrate and affect soil biota; formaldehyde degrades but can be acutely toxic at high concentrations.

D. Fertilizer plants (BVFCL Namrup; similar ammonia-urea complexes)

Environmental effects:

- **Air:** Fugitive **ammonia (NH_3)** emissions, NO_x from combustion, particulate matter from prilling/handling.frontiersin+1
- **Water:** Ammonia/urea leaks can cause **eutrophication**, oxygen depletion, and toxicity to aquatic life.
- **Soil:** High nitrogen deposition can alter soil chemistry and microbial communities.
- **Accidents:** Ammonia leaks cause acute toxicity to humans and animals; large releases can form toxic plumes extending **hundreds of meters to a few kilometers**, depending on conditions.ukpandi+2

E. Petrochemical cracker & polymer plant (BCPL, Lepetkata)

Environmental effects:

- **Air:** VOCs, olefins (ethylene, propylene), PM, NO_x , SO_x from furnaces; potential benzene and other aromatics.[ammoniagas](#)
- **Water:** Process wastewater with organics, oils, traces of monomers.

- **Soil:** Spills of hydrocarbons/monomers can contaminate soil.
- **Major accident risk:** Vapor cloud explosions and large fires can produce extensive thermal radiation and toxic plumes.

F. Chemical traders / small manufacturers (acids, solvents, bleaching powder, etc.)

Environmental effects:

- **Air:** Solvent vapors (VOCs), acid mists, chlorine gas if bleach mixes with acids.
- **Water/soil:** Spills of acids, solvents, bleaching agents into drains/soil can harm aquatic life, corrode infrastructure, and contaminate groundwater.
- **Warehouse fires:** Mixed chemical fires generate **complex toxic smoke** (HCl, SO₂, NO_x, organics) and contaminated runoff.

G. Chemical warehouse / logistics hubs

Environmental effects:

- **Air:** Smoke from fires containing a mix of organics, acids, particulates.
- **Water/soil:** Firefighting water carrying dissolved chemicals into storm drains, rivers, and soil.
- **Risk:** Cascading reactions if incompatible chemicals are stored together.

Through which it can spread into environment :air, water, soil, and biota.

Airborne dispersion (most critical for emergencies)

- **Gases & vapors:** Ammonia, methanol, formaldehyde, VOCs, H₂S, SO₂, NO_x, chlorine, etc. disperse as **plumes** carried by wind.gem+2
- **Particulates:** PM_{2.5}/PM₁₀, soot, aerosols from fires or process emissions travel with wind and settle over time.
- **Thermal plumes:** Fires create buoyant plumes that rise, then spread downwind; thermal radiation affects areas in line-of-sight.gem+1

Key drivers:

- **Wind speed & direction** (dominant factor)
- **Atmospheric stability** (how much vertical mixing occurs)
- **Terrain & buildings** (urban roughness, valleys, river corridors)
- **Release height & duration** (ground-level vs elevated stack; instantaneous vs continuous)

Waterborne spread

- **Direct spills** into drains/nallahs → flow into **Brahmaputra** or local rivers (e.g., formalin into Jatinga River).indiatodayne+2
- **Firefighting runoff** from refineries/warehouses carries hydrocarbons, acids, solvents into water bodies.
- **Leaching** from contaminated soil into groundwater over time.

Soil & groundwater

- **Infiltration** of spilled liquids (hydrocarbons, solvents, acids) into soil.
- Long-term **groundwater plumes** moving with hydraulic gradient, potentially affecting wells.

How far contamination can spread (order-of-magnitude distances)

These are **typical ranges** for emergency planning, not exact predictions.

A. Refinery (IOCL Guwahati)

- **Routine emissions:** Elevated pollutant concentrations typically within **5–10 km**, strongest within **2–5 km** downwind; some epidemiological signals seen up to **20 km**.[pmc.ncbi.nlm.nih](https://pubmed.ncbi.nlm.nih.gov/16111111/)
- **Major accident (tank fire, large vapor release):**
 - **Thermal radiation** from large pool fires can cause severe burns and structural damage out to **100–300 m**, sometimes more for very large tanks.gem+1
 - **Toxic plume** (e.g., H₂S, VOCs) can extend **1–3 km** for serious releases, depending on wind and stability.

B. Industrial gas plants

- **Cylinder/cascade rupture (acetylene, oxygen):**
 - Blast/fragment hazard: **tens of meters**.
 - Fire/thermal radiation: **50–150 m** for significant fires.
- **Large bulk tank failure:** Toxic/asphyxiant cloud could extend **hundreds of meters to ~1 km** downwind in worst cases.

C. Methanol / formalin facilities & tanker spills

- **Airborne methanol/formaldehyde vapor cloud:**
 - Hazardous concentrations typically **hundreds of meters to ~1–2 km** downwind for large spills/fires.
- **River spill (formalin):**

- Contaminated stretch can extend **several kilometers downstream**; in the Jatinga incident, authorities banned water use along the river for **48 hours** and monitored downstream areas.indiatodayne+2

D. Ammonia-urea fertilizer plants (BVFCL-type)

- **Small cylinder/line leak:** Initial isolation **30–60 m**; downwind protection distances **100–800 m** depending on size and time of day.ukpandi+1
- **Large storage/tank failure:**
 - Initial isolation **300+ m**; downwind protective distances can reach **1–2 km** or more at night under stable conditions.ukpandi+1
- **Water/soil:** Localized around the plant; ammonia dissolves quickly in water but can affect aquatic life over **hundreds of meters to a few kilometers** of stream/river if large quantities enter.

E. Petrochemical cracker (BCPL-type)

- **Vapor cloud explosion / large hydrocarbon release:**
 - Thermal radiation and overpressure effects: **hundreds of meters** for severe damage;
 - Toxic plume (olefins, aromatics): potentially **1–3 km** downwind for major incidents.
- **Routine VOC dispersion:** Elevated concentrations within **1–5 km**, especially near storage tanks and wastewater units.[ammoniagas](#)

F. Small chemical units & warehouses

- **Godown fire with mixed chemicals:**
 - Smoke plume with irritants/toxins: **1–5 km** downwind depending on fire size and meteorology.
 - Runoff contamination: local drains and nearby water bodies within **hundreds of meters to a few kilometers**.

we can predict these things :

Concentration fields (e.g., ppm or mg/m³) over a grid around the facility at specific time

Hazard zone boundaries:

- AEGL/ERPG exceedance distances (e.g., where concentration > threshold for mild/moderate/severe effects)
- Flammable vapor cloud extent
- Thermal radiation levels (kW/m²) for firesgem+1

Impact metrics:

- Number of people exposed above certain thresholds

- Area of land/water affected
- Time until concentration falls below safe levels at key locations

Can we get live wind and air quality data?

Yes, at least for **near-real-time** inputs to your model.

Wind & meteorological data

Possible sources:

- **India Meteorological Department (IMD):**
 - Provides station data (wind speed/direction, temperature, etc.) for major cities including Guwahati.
 - Some datasets are available via Python packages (e.g., `imddata` for gridded rainfall/temp; wind may require direct IMD or third-party access).[missrifka](#)
- **Central Water Commission (CWC) telemetry:**
 - Hourly **wind speed** data from meteorological stations, accessible as CSV/API.[scribd](#)
- **OpenWeather / Windy / other global APIs:**
 - Offer current wind speed/direction and forecasts for Guwahati coordinates; useful for MVP even if not official Indian data.

How wind affects chemical spread

- **Direction:** Determines **where** the plume goes; areas directly downwind face highest concentrations.
- **Speed:**
 - Low wind → plume stays concentrated, travels slower, can cause high local concentrations.
 - High wind → plume dilutes faster but reaches farther downwind quickly.

note : A **plume** is the cloud-like shape that a released substance (gas, smoke, particles, or even heat) forms as it moves away from its source and spreads through the surrounding environment, usually air or water.[knowledge.gexcon+1](#)